04/08/2022

Friday, April 8, 2022 10:50 AM

WIR9

Mr. Orchard's Math 141 WIR Test 3 Review Week 11

1. The frequency distribution of the hourly wage rate (in dollars) of workers at a certain factory is given in the table below.

Wage Rate	\$10.30	\$10.40	\$10.50	\$10.60	\$10.70	\$10.80
Frequency	60	90	75	120	60	45

Find the mean, median, mode, standard deviation and variance of the data.

1 | 10.90

120

2. A game store manager knows that the probability that a videogame will be returned to the store is 0.22. If 63 games were sold in a week, determine the probabilities of the following events. (Round your answers to two decimal places.)

   (a) No more than 20 games will be returned.

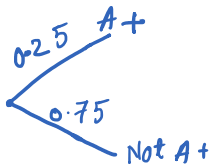
   (b) At least 9 games will be returned.

   (c) More than 6, but fewer than 15 games will be returned.

   (d) How many games should the store manager expect to have returned?

   (e) What is the standard deviation of the number of games returned?

3. It is estimated that one fourth of the general population has blood type A+. A sample of six people is selected at random. What is the probability that exactly three of them have blood type A+? Round your answer to four decimal places.



4. The probability distribution of a random variable X is given below:

x	1	2	3	4	5	6	7	8
$P(X = x)$	0.1	0.15	0.15	0.05	0.1	0.1	0.2	0.15

Find the expected value and the variance of the random variable. Round your answer to two decimal places.

5. (a) The probability of catching a certain pokémon is 0.3. What are the odds against catching that pokémon?

(b) The odds of catching a different pokémon are 4 to 9 in favor. What is the probability of catching this pokémon?

6. Classify the following random variables and give the range of values that the random variable may take.

   (a) 1000 werewolves are locked in cages the night of a full moon. After they transform, we record the number of times each werewolf howls at the moon before transforming back to a human. X = the number of times a werewolf howls at the moon before transforming back to a human.

   Infinite Discrete

   (b) In a bag of Bertie Bott's Every Flavor Beans, there are 6 old oak door flavored beans and 15 mead flavored beans. A bean is drawn at random from the bag and not replaced. Y = the number of draws it takes to get an old oak door flavored bean is picked.

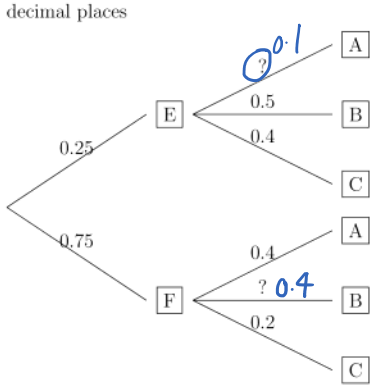
   F. D.

   (c) A cat toy is placed on the ground. 12 cats are given a chance to play with the toy and the length of time they play is recorded in a random variable, Z.

   Continuous.

7. A student studying for a vocabulary test knows the meanings of 12 words from a list of 20. If the test contains 10 of the words from the list, what is the probability that at least 8 of the words on the test are words that the student knows? (Round to 3 decimal places.)

8. Use the tree diagram to answer the following questions. Round your answers to three decimal places



(a) $P(E|A) = \dfrac{P(E\cap A)}{P(E\cap A) + P(F\cap A)} = \dfrac{0.25 \times 0.1}{0.25 \times 0.1 + 0.75 \times 0.4}$

$= \dfrac{0.025}{0.025 + 0.3}$

(b) $P(A|E)$

(c) $P(C) = P(E\cap C) + P(F\cap C)$

$= 0.25 \times 0.4 + 0.75 \times 0.2 = 0.077$

(d) $P(B\cap E)$

$= 0.1 + 0.15$

$= 0.25$

9. Let $S = \{s_1,\ s_2,\ s_3,\ s_4,\ s_5,\ s_6\}$ be the sample space associated with an experiment having the following probability distribution.

Outcome	s_1	s_2	s_3	s_4	s_5	s_6
Probability	$\frac{2}{34}$		$\frac{6}{34}$	$\frac{7}{34}$	$\frac{6}{34}$	$\frac{9}{34}$

Calculate the following probabilities.

(a) $P(\{s_2\})$

(b) $P(\{s_1,\ s_2,\ s_3\})$

10. At a taco food truck, you can order one, two, or three tacos. With each of those choices, you have the choice of a small drink, a large drink, or no drink. The information for patrons at the food truck last Friday is given in the table below.

	Small Drink	**Large Drink**	**No Drink**	**Total**
One Taco	6	8	2	16
Two Tacos	0	40	6	46
Three Tacos	3	15	20	38
Total	9	63	28	100

(a) What is the probability that a person ordered a large drink given that they ordered a 3 taco meal?

(b) What is the probability that a person ordered one taco and got a small drink?

(c) What is the probability that a person who has a small drink ordered a 2 taco meal?

11. Let E and F be two independent events, and suppose $P(E) = 0.4$ and $P(F) = 0.1$. Compute the probabilities below.
    (a) $P(E \cap F)$

    (b) $P(E \cup F)$

    (c) $P(E^c)$

12. Let E and F be two mutually exclusive events, and suppose $P(E) = 0.2$ and $P(F) = 0.5$. Compute the probabilities below.
    (a) $P(E \cap F)$

    (b) $P(E \cup F)$

    (c) $P(E^c)$

13. Four cards are selected at random without replacement from a well-shuffled deck of 52 playing cards. Find the probabilities of the given events. Round your answers to four decimal places.

    (a) All four cards are spades.

    (b) All four cards are of the same suit.

    (c) No face cards (jack, queen, or king) are drawn.

    (d) All four cards are face cards.